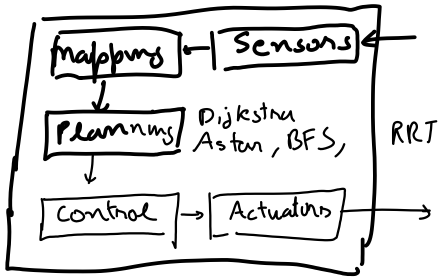
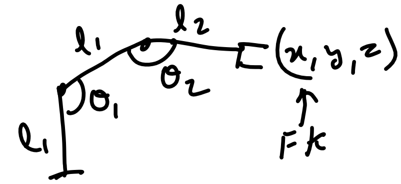
Control

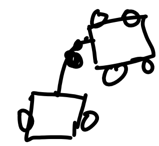
Mapping



Robotic arm

R, T → Forward Kinematics (FK)
→ Inverse Kinematics (IK)



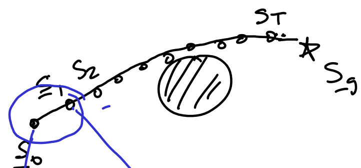


MEE 444
Robotic control

Planning : Going from start state to goal state
↓
Trajectory in terms of states

Higher-level of abstraction

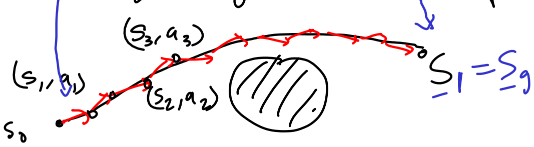
Long trajectories



Control : Start state → goal state
↓ soln
Trajectory in terms of state-action pairs / policy function
state feedback control

Lower level of abstraction

follow the trajectory provided by the Planner



Control
- PID : Proportional - Integral - Differential controller (P, PI, PD)
- LQR : Linear Quadratic Regulator → iLQR incremental LQR
- MPC : Model Predictive control
- RL : Reinforcement Learning → Q-learning (Tabular form)

PID controller

single dimensional control

for multi dimensional control use one PID per dimension

For Jetbot: what is the state space, action space, system dynamics, and cost function?

We only covered Discrete planning

But Jetbot has states and action in cont. domain.

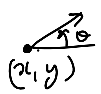
Simplified mathematical models for robot like Jetbots

↳ Unicycle model ✓

↳ Ackermann drive

$$\text{state} = \begin{bmatrix} x \\ y \\ \theta \end{bmatrix} = \underline{s}_t \qquad \underline{s}_t \in S$$

S ↑ all possible value of state = state space



$$\underline{a}_t = \text{action} = \begin{bmatrix} v_t \\ \omega_t \end{bmatrix}$$

v_t — linear velocity

ω_t — angular velocity

action ↑ control signal (u_t)

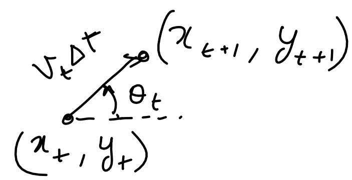
$a_t \in A$ ↑ Action space

$\underline{u}_t$ RL $\underline{a}_t$

System dynamics (discrete time, Δt)

$$\underline{s}_{t+1} = f(\underline{s}_t, \underline{a}_t)$$

$$\underbrace{\begin{bmatrix} x_{t+1} \\ y_{t+1} \\ \theta_{t+1} \end{bmatrix}}_{\underline{s}_{t+1}} = \underbrace{\begin{bmatrix} x_t + v_t \Delta t \cos\theta_t \\ y_t + v_t \Delta t \sin\theta_t \\ \theta_t + \omega_t \Delta t \end{bmatrix}}_{f(\underline{s}_t, \underline{a}_t)}$$



Acceleration

$$\underline{s}_t = \begin{bmatrix} x_t \\ y_t \\ v_t \\ \theta_t \end{bmatrix} \qquad \underline{a}_t = \begin{bmatrix} b_t \\ \omega_t \end{bmatrix}$$

b_t linear acc

ω_t ang. vel

Markovian state

must capture all the info that is required to predict next state

$$\underline{s}_{t+1} \perp s_{t-1} \mid \underline{s}_t$$

s_{t+1} is conditionally ind. of s_{t-1} given $\underline{s}_t$

cost function = defines the task

$$c(\underline{s}_t, \underline{a}_t) = \|\underline{s}_t - \underline{s}_g\|^2 + \lambda \|\underline{a}_t\|^2 \quad \leftarrow \text{example}$$

<u>Control problem</u>

$$\{\underline{a}_t\}_{t=1}^{T} = \underset{\{\underline{a}_t\}_{t=1}^{T}}{\text{minimize}} \sum_{t=1}^{T} c(s_t, a_t)$$

$$\text{subject to} \quad \underline{s}_{t+1} = f(\underline{s}_t, a_t) \quad \forall t \in \{1, \dots T\}$$

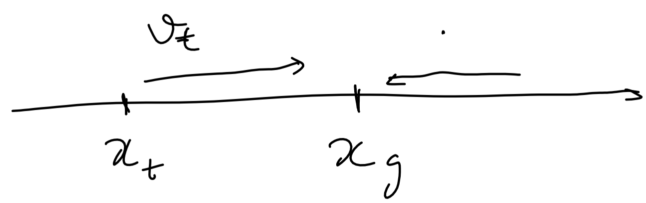
$$\pi^*(\underline{s}_t) = \underset{\pi}{\text{minimize}} \sum_{k=t}^{T} c(\underline{s}_k, \pi(\underline{s}_k))$$

policy function

$$\text{subject to} \quad \underline{s}_{k+1} = f(\underline{s}_k, \pi(\underline{s}_k)) \quad \forall k \in \{t, \dots T\}$$

PID controller does not use system dynamics or the cost function. Hence it is not called optimal control

We construct an error function (must be signed



$$e(x_t) = x_g - x_t$$

$$\underline{a}_t = \underbrace{k_p e(x_t)}_{\text{Proportional Term}} + \underbrace{k_I \int_0^t e(x_t)\,dt}_{\text{Integral}} + \underbrace{k_d \cdot \frac{d}{dt} e(x_t)}_{\text{Differential term}}$$

Linear Quadratic Control/Regulator (LQR)

$$\{a_t\}_{t=1}^{T} = \underset{\{a\}}{\arg\min} \sum \text{cost function}$$

Quadratic in state and action

subject to $\underbrace{\text{system dynamics}}$

Linear in state and action

$$c(\underline{s}_t, \underline{a}_t) = \underline{s}_t^T Q_t \underline{s}_t + \underline{a}_t^T R_t \underline{a}_t + \underline{s}_t^T N \underline{a}_t + \underline{q}^T \underline{s}_t + \underline{r}^T \underline{a}_t$$

Using homogeneous coordinates linear terms like $\underline{q}^T \underline{s}_t$ can be absorbed in the quadratic term.

$$\tilde{\underline{s}}_t = \begin{bmatrix} \underline{s}_t \\ 1 \end{bmatrix} \Rightarrow \underline{s}_t^T Q \underline{s}_t + \underline{q}^T \underline{s}_t \stackrel{?}{=} \tilde{\underline{s}}_t^T \tilde{Q}_t \tilde{\underline{s}}_t$$

$$= [\underline{s}_t^T \quad 1] \begin{bmatrix} Q & x \\ x^T & 1 \end{bmatrix} \begin{bmatrix} \underline{s}_t \\ 1 \end{bmatrix}$$

$$= [\underline{s}_t^T \quad 1] \begin{bmatrix} Q\underline{s}_t + \underline{x} \\ \underline{x}^T s_t + 1 \end{bmatrix} \longrightarrow = \underline{s}_t^T Q \underline{s}_t + \underline{s}_t^T \underline{x} + \underline{x}^T \underline{s}_t + 1$$

$\in \mathbb{R}$

$$= \underline{s}_t^T Q \underline{s}_t + \underline{s}_t^T \underline{x} + \underline{x}^T \underline{s}_t + \underline{1}$$

$$(s_t^T \underline{x})^T = \underline{x}^T \underline{s}_t = \underline{x}^T \underline{s}_t$$

$$\underline{s}_t^T Q \underline{s}_t + \underline{q}^T \underline{s}_t = \underline{s}_t^T Q \underline{s}_t + 2\underline{x}^T \underline{s}_t + 1$$

$$\Rightarrow 2\underline{x} = \underline{q} \Rightarrow \underline{x} = \underline{q}/2$$

$$\underbrace{\underline{s}_t^T Q \underline{s}_t}_{\text{quad}} + \underbrace{\underline{q}^T \underline{s}_t}_{\text{linear}} = \underbrace{\underline{\tilde{s}}_t^T \hat{Q}}_{\text{quad}} \underline{\tilde{s}}_t - 1$$

homogeneous coordinates

$$\hat{\underline{Q}} = \begin{bmatrix} Q & \underline{q}/2 \\ \underline{q}^T/2 & 1 \end{bmatrix} \qquad \tilde{s}_t = \begin{bmatrix} \underline{s}_t \\ 1 \end{bmatrix}$$

LQR:

$$\underset{\{a_t\}_{t=1}^T}{\text{minimize}} \overbrace{\sum_{t=1}^{T} \underline{s}_t^T Q_t \underline{s}_t + \underline{a}_t^T R_t \underline{a}_t}$$

$$\text{subject to } \underbrace{\underline{s}_{t+1} = A\underline{s}_t + B\underline{a}_t}$$

$$\underline{s}_{t+1} = f(\underline{s}_t, \underline{a}_t)$$

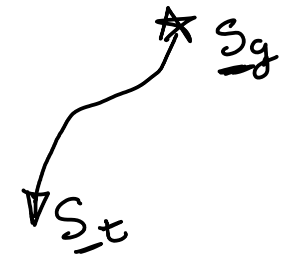
$$\underline{s}_1 \xrightarrow{\underline{a}_1} \underline{s}_2 \; \cdots\cdots \; \rightarrow \underline{s}_{T-1} \xrightarrow{\underline{a}_{T-1}} \underline{s}_T$$

LQR:

$$\underset{\{\underline{a}_t\}_{t=1}^{T-1}}{\text{minimize}} \left(\sum_{t=1}^{T-1} \underline{s}_t^T Q_t \underline{s}_t + \underline{a}_t^T R_t \underline{a}_t \right) + \left(\underline{s}_T^T Q_T \underline{s}_T \right)$$

$$\text{s.t.} \quad \underline{s}_{t+1} = A_t \underline{s}_t + B_t \underline{a}_t \quad \forall \; t = 1, \dots, T-1$$

Example:



$$\|\underline{s}_T - \underline{s}_g\|_2^2 \leftarrow \text{quadratic cost}$$

$$= (\underline{s}_T - \underline{s}_g)^T (s_T - s_g)$$

$$= \underbrace{\underline{s}_T^T \underline{s}_T}_{\text{quad}} - 2 \underbrace{\underline{s}_g^T s_T}_{\text{linear}} + \underline{s}_g^T s_g$$

$$= \underbrace{\begin{bmatrix} s_T^T & 1 \end{bmatrix}}_{\tilde{\underline{s}}_T^T} \underbrace{\begin{bmatrix} I_{n \times n} & -\underline{s}_g \\ \underline{s}_g^T & \underline{s}_g^T s_g \end{bmatrix}}_{\tilde{Q}_T} \underbrace{\begin{bmatrix} \underline{s}_T \\ 1 \end{bmatrix}}_{\tilde{\underline{s}}_T} = \tilde{\underline{s}}_T^T \tilde{Q}_T \tilde{\underline{s}}_T$$

Expressed in the LQR cost function form

Trade off between energy spent and going to goal

$$\text{cost} = \|\underline{s}_t - \underline{s}_g\|_2^2 + \frac{1}{2} m v_t^2 + \frac{1}{2} I \omega_t^2$$

$$\underline{a}_t = \begin{bmatrix} v_t \\ \omega_t \end{bmatrix}$$

$$= \tilde{\underline{S}}_t^T Q_t \tilde{\underline{S}}_t + \underbrace{[v_t \ \ \omega_t]^T}_{\vec{\underline{a}}_t} \underbrace{\begin{bmatrix} \frac{1}{2}m & 0 \\ 0 & \frac{1}{2}I \end{bmatrix}}_{R.} \underbrace{\begin{bmatrix} v_t \\ \omega_t \end{bmatrix}}_{\underline{a}_t}$$

Q_t matrix captures preferences over states

R_t matrix typically penalizes big actions and leads to smoother trajectories

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Unicycle system dynamics is not Linear
non-Linear

$$\begin{bmatrix} x_{t+1} \\ y_{t+1} \\ \theta_{t+1} \end{bmatrix} = \begin{bmatrix} x_t + v_t \underline{\cos\theta_t} \Delta t \\ y_t + v_t \underline{\sin\theta_t} \Delta t \\ \theta_t + \omega_t \Delta t \end{bmatrix} \qquad \underline{s}_{t+1} = A\underline{s}_t + B\underline{a}_t$$

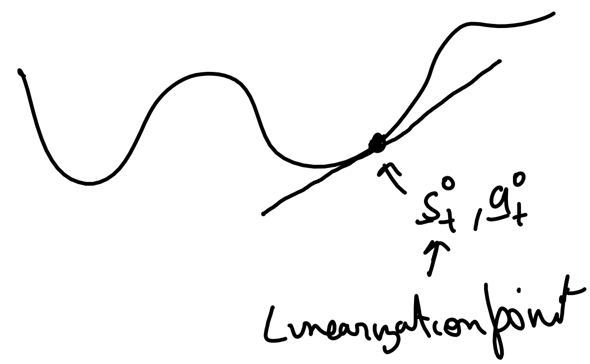
which terms are non-Linear?

Try to make linear

$$\begin{bmatrix} x_{t+1} \\ y_{t+1} \\ \cos\theta_{t+1} \\ \sin\theta_{t+1} \end{bmatrix} = \begin{bmatrix} x_t + v_t \underline{\cos\theta_t} \Delta t \\ y_t + v_t \underline{\sin\theta_t} \Delta t \\ \cos\theta_t + \_\_\_\_ \\ \sin\theta_t + \_\_\_\_ \end{bmatrix}$$

Is it Linear now?, If not due to which Terms?

Taylor series is used for local linear approx.



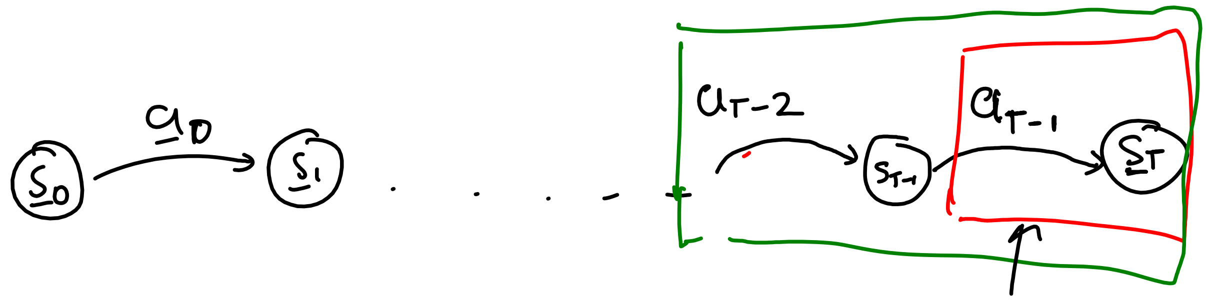
$$\underline{s}_{t+1} = \underline{f}(\underline{s}_t, \underline{a}_t)$$

Linearize around $\underline{s}_t^0, \underline{a}_t^0$

$$\underline{f}(\underline{s}_t, \underline{a}_t) \approx \underline{f}(\underline{s}_t^0, \underline{a}_t^0) + \left[J_{\underline{s}_t} f\right](\underline{s}_t - \underline{s}_t^0) + \left[J_{\underline{a}_t} f\right](\underline{a}_t - \underline{a}_t^0)$$

Jacobian matrices

$$\approx \underline{f}(\underline{s}_t^0, \underline{a}_t^0) + \frac{\partial \underline{f}}{\partial \underline{s}_t}(\underline{s}_t - \underline{s}_t^0) + \frac{\partial \underline{f}}{\partial \underline{a}_t}(\underline{a}_t - \underline{a}_t^0)$$



LQR:

$$\underset{\{a_t\}_{t=1}^{T-1}}{\text{minimize}} \quad \left(\sum_{t=1}^{T-1} \underline{s}_t^T Q_t \underline{s}_t + \underline{a}_t^T R_t \underline{a}_t\right) + \underline{s}_T^T Q_T \underline{s}_T$$

$$\text{s.t.} \quad \underline{s}_{t+1} = A_t \underline{s}_t + B_t \underline{a}_t \quad \forall \; t = 1, \dots, T-1$$

$$\underset{a_{T-1}}{\text{minimize}} \quad \underline{a}_{T-1}^T R \, \underline{a}_{T-1} + \underline{s}_T^T Q \, \underline{s}_T$$

$$\text{s.t} \quad \underline{s}_T = A \underline{s}_{T-1} + B \underline{a}_{T-1}$$

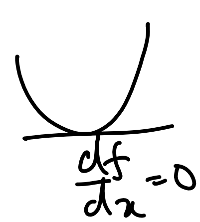
$$\equiv \quad \underset{a_{T-1}}{\text{minimize}} \quad \underline{a}_{T-1}^T R \, \underline{a}_{T+1} + \left(A \underline{s}_{T-1} + B \underline{a}_{T-1}\right)^T Q \left(A \underline{s}_{T-1} + B \underline{a}_{T-1}\right)$$

How to minimize a Quadratic vector function?

For scalar function:

$$f(x) = ax^2 + bx + c \qquad \frac{df}{dx} = 0$$

$$f(x) = a\left(x^2 + \frac{b}{a} x\right) + c \qquad \text{if } a > 0$$



$$= a\left(x^2 + \frac{b}{a} x + \frac{b^2}{4a^2} - \frac{b^2}{4a^2}\right) + c$$

$$f(x) = \underbrace{a\left(x+\frac{b}{2a}\right)^2}_{0} + \underbrace{c - \frac{b^2}{4a^2}} \qquad \text{if } a>0$$

$f(x)$ is minimum at $x = -\frac{b}{2a}$

Can we do something similar for vector function

$$f(\underline{x}) = \underset{\text{vector}}{\underline{x}^T} \underset{\text{matrix}}{A} \underline{x} + \underline{b}^T \underline{x} + c$$

$$\frac{\partial}{\partial \underline{x}} \underline{x}^T A \underline{x} = 2\underline{x}^T A$$

$$\frac{\partial}{\partial \underline{x}} \underline{b}^T \underline{x} = \underline{b}^T$$

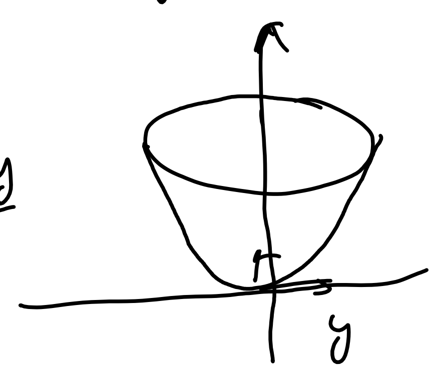
$a > 0$,

$A \succ 0$? Positive definiteness (PD)

A matrix A is PD if $\underline{y}^T A \underline{y} > 0$ for all $\underline{y} \in \mathbb{R}$

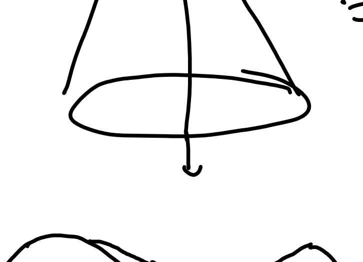
PD $\equiv$ all eigen values are +ve

$\underline{y}^T A \underline{y}$



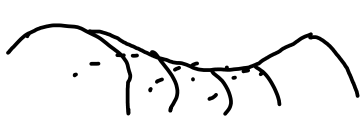
negative definite

$\underline{y}^T A \underline{y} < 0$



$\equiv$ all eigen values are −ve

saddle point



$\equiv$ some +ve, some −ve

$$\underline{y}^T A \underline{y} = \lambda \underbrace{\underline{y}^T \underline{y}} =$$

eigen vectors = unit vector

$$\underline{y}^T A \underline{y} = \lambda \|y\|^2$$

$$\frac{\underline{y}^T}{\|y\|} A \frac{\underline{y}}{\|y\|} = \lambda \longleftarrow \text{eigen value}$$

if $\underline{y}^T A \underline{y} > 0 \longrightarrow$ eigen value are +ve

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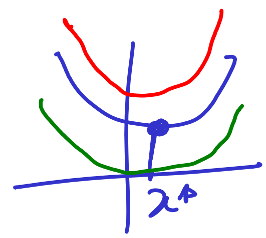
$$x^* = \arg\min_{\underline{x}} f(\underline{x})$$



$$f(\underline{x}) = \underline{x}^T A \underline{x} + \underline{b}^T \underline{x} + \underbrace{c}_{\text{independent}}$$

$$\downarrow$$

$$(M\underline{x} + \underline{n})^T (M\underline{x} + \underline{n}) + p$$

$$= ((M\underline{x})^T + \underline{n}^T)(M\underline{x} + n) + p$$

$$= (\underline{x}^T M^T + \underline{n}^T)(M\underline{x} + n) + p$$

$$= \underline{x}^T M^T M \underline{x} + \underline{x}^T M^T \underline{n} + \underline{n}^T M \underline{x} + \underline{n}^T \underline{n} + p$$

$$(A+B)^T = A^T + B^T$$

$$(A \cdot B)^T = B^T A^T$$

$$(\underline{n}^T M \underline{x})^T = \underline{x}^T M^T \underline{n}$$

$$= \underline{x}^T M^T M \underline{x} + 2\underline{n}^T M \underline{x} + \underbrace{\underline{n}^T \underline{n} + p}_{\text{independent of } \underline{x}}$$

A is P.D. $\Leftarrow \underbrace{M^T M = A}$, $2n^T M = \underline{b}^T \Rightarrow M^T \underline{n} = \frac{\underline{b}}{2} \Rightarrow \underline{n} = M^{-T} \frac{\underline{b}}{2}$

$$f(\underline{x}) = \overbrace{(M\underline{x}+\underline{n})^T(M\underline{x}+\underline{n})}^{=0} + p$$

assume $A > 0$ A is Positive definite / invertible

$$\underline{x}^T A \underline{x} = \underline{x}^T M^T M \underline{x} = (M\underline{x})^T (M\underline{x})$$

$$= \|M\underline{x}\|^2 > 0$$

whether M is invertible ?

assume M is "

$$\underline{x}^* = -M^{-1}\underline{n}$$

$$M\underline{x} + \underline{n} = 0$$

$$x = -M^{-1}\underline{n}$$

$$\underline{x}^* = -M^{-1}\left(M^{-T}\frac{\underline{b}}{2}\right)$$

$$= -\underbrace{(M^T M)}^{-1}\frac{\underline{b}}{2}$$

$$(AB)^{-1} = B^{-1}A^{-1}$$

$$\boxed{\underline{x}^* = -\frac{A^{-1}\underline{b}}{2}}$$ is The minimum of

$$f(\underline{x}) = \underline{x}^T A \underline{x} + \underline{b}^T \underline{x} + c$$

LQR last step

$$\underset{a_{T-1}}{\text{minimize}}\ \underline{a}_{T-1}^T R \underline{a}_{T+1} + (A\underline{s}_{T-1} + B\underline{a}_{T-1})^T Q (A\underline{s}_{T-1} + B\underline{a}_{T-1})$$

$a^*_{T-1} = ?$ find the minima using above solution

$$= \underline{a}_{T-1}^T R \underline{a}_{T-1} + \left[(A\underline{s}_{T-1})^T + (B\underline{a}_{T-1})^T\right] Q (A\underline{s}_{T-1} + B\underline{a}_{T-1})$$

$$= \underline{a}_{T-1}^T R \underline{a}_{T-1} + (\underline{s}_{T-1}^T A^T + \underline{a}_{T-1}^T B^T) Q (A\underline{s}_{T-1} + B\underline{a}_{T-1})$$

$$= \underline{a}_{T-1}^T R \underline{a}_{T-1} + \underline{s}_{T-1}^T A^T Q A \underline{s}_{T-1} + \underline{a}_{T-1}^T B^T Q B \underline{a}_{T-1}$$

$$+ \underline{s}_{T-1}^T A^T Q B \underline{a}_{T-1} + \underline{a}_{T-1}^T B^T Q A \underline{s}_{T-1}$$

$$= \underline{a}_{T-1}^T \underbrace{(R + B^T Q B)}_{A} \underline{a}_{T-1} + \underbrace{2\underline{s}_{T-1}^T A^T Q B}_{b^T} a_{T-1}$$

$$+ \underline{s}_{T-1}^T A^T Q A \underline{s}_{T-1}$$

$$\underline{a}_{T-1}^* = -\frac{1}{2} A^{-1} \underline{b}$$

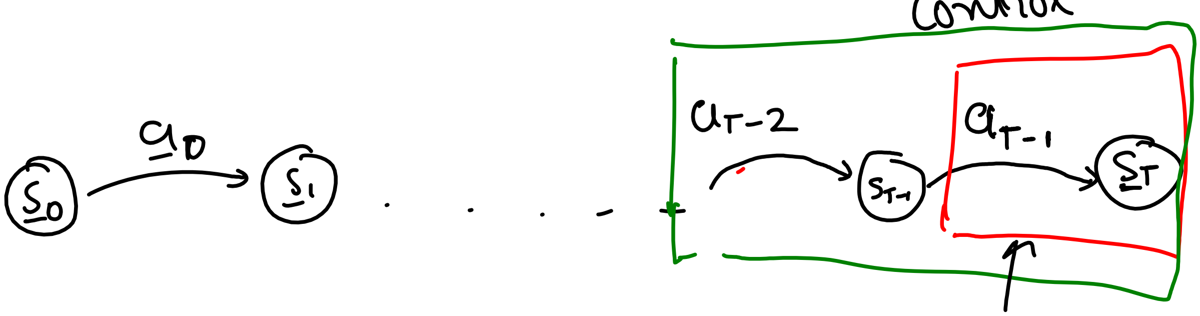
$$= -\frac{1}{\cancel{2}} (R + B^T Q B)^{-1} (B^T Q A \underline{s}_{T-1}) \cancel{2}$$

$$= -\underbrace{(R + B^T Q B)^{-1} B^T Q A}_{\text{feedback}} \underline{s}_{T-1}$$

$$K_{T-1}$$

$$\underline{a}_{T-1}^* = -K_{T-1} \underline{s}_{T-1}$$

Linear feedback control



$$\underset{a_{T-2}}{\text{minimize}} \quad a_{T-2}^T R a_{T-2} + \underbrace{\quad\quad\quad\quad}_{\text{using } a^*_{T-1}}$$

cost going forward

cost-to-go

$$\underbrace{\underline{s}_{T-1}^T P \, s_{T-1}}$$

Put back optimal $\underline{a}^*_{T-1}$ in

$$\underline{a}_{T-1}^T R \underline{a}_{T+1} + (A\underline{s}_{T-1} + B\underline{a}_{T-1}) \underline{Q} (A\underline{s}_{T-1} + B\underline{a}_{T-1})$$

$$= s_{T-1}^T P s_{T-1}$$